

JAYESH J. PANDEY

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student (GPA 8.70) with expertise in Machine Learning, Deep Learning, and LLM applications. NASA Space Apps Global Winner experienced in building ML pipelines, RAG systems, multi-agent frameworks, and satellite data analysis. Maintainer of 89+ GitHub repositories spanning AI, space-tech, and computer vision.

EDUCATION

PVGOE SSDIOM, Nashik Aug 2022 – June 2026
B.E. Computer Engineering | GPA: 8.70

PROFESSIONAL EXPERIENCE

AI/ML Intern May 2026 – Present
PGAGI Remote

- Developing and optimizing AI/ML models for production deployment and research initiatives.

Research Intern May 2026 – Present
IEEE Remote

- Conducting research in ML/AI domains; contributing to publications and technical documentation.

Project Trainee – Machine Learning Feb 2026 – Present
Physical Research Laboratory (PRL) / ISRO Ahmedabad, India

- Developing PyTorch-based neural networks for semi-Gaussian signal parameter identification and real-time classification using low-sampling rate data.
- Optimizing deep learning architectures and integrating IoT pipelines for AI-powered space weather monitoring and satellite signal analysis.

LLMOps Engineer Intern Aug 2025 – Jan 2026
EaseMeMed Remote

- Built and optimized PyTorch object detection models using transfer learning, data augmentation, and MLOps best practices.
- Implemented LLM inference optimization using quantization and vLLM, significantly reducing latency and resource consumption.

Freelance Developer & AI Consultant 2023 – Present
Independent Remote

- Delivered end-to-end ML projects for engineering students covering ML pipelines, computer vision, NLP, and web/mobile apps.
- Built AI-powered solutions for small businesses: automated appointment systems, chatbots, and management tools using FastAPI, React, and LLM agents.

Open Source Contributor Aug 2024 – Nov 2025
GirlScript Summer of Code (GSSOC) Remote

- Contributed to multiple open-source repositories in Natural Language Processing and Computer Vision domains.

SKILLS

Programming & Databases: Python, C++, SQL, PostgreSQL, MySQL, MongoDB

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Scikit-learn, Statistical Analysis, EDA, A/B Testing

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CNN, LSTM, Transfer Learning, Transformer Architecture, Fine-tuning (LoRA / QLoRA), Attention Mechanisms, XGBoost, CatBoost, Reward Modeling, RLHF

LLM & Generative AI: Large Language Models, RAG, LangChain, LangGraph, Llama 3.2, Quantization, vLLM, Multi-Agent Systems, Prompt Engineering, Vector Databases (FAISS / Chroma), Sentence Embeddings, Semantic Search, Bit-Quantization, StyleGAN2

Computer Vision & Geospatial: GDAL, Sentinel-2, Remote Sensing, Astropy, Image Segmentation, Object Detection, Feature Pyramid Networks, Multimodal AI, CLIP, ImageReward

MLOps & Cloud: AWS (EC2 / S3), Docker, Git, MLflow, CI/CD Pipelines, Model Deployment, FastAPI, Flask, Linux

Tools & Technologies: React Native, Neo4j, Claude-Code, Opencode, Ollama, Weights & Biases

PROJECTS

Ekip Bhaskar – Halo CME Early Warning System | [GitHub](#)

NASA 2025 – Global Winner

- Developed AI model for Halo CME detection using Aditya-L1 SWIS-ASPEX satellite data achieving 84% accuracy.
- Tech: Python, Pandas, SciPy, Astropy, Matplotlib, React Native.

Graph-Based Intelligent RAG System | [GitHub](#)

Full-Stack LLM App

- Built Neo4j knowledge graph with LangGraph agents for natural language querying over business documents with interactive visualizations.
- Tech: Neo4j, LangGraph, FastAPI, React, Llama 3.2, Cytoscape.js.

VLMJudge – Multimodal Evaluation Framework

VLM / RLHF / Reward Modeling

- Production-grade multimodal evaluation framework for image comparison and preference learning with 89.7% accuracy and 320ms latency.
- Hybrid student-teacher architecture with CLIP, Qwen2.5-VL ensemble, confidence calibration, and continuous learning pipeline.

L3AGI – Team of AI Agents | [GitHub](#)

Open-Source Multi-Agent

- Open-source framework for collaborative AI agent teams with memory, toolkits, data integrations, and report generation.
- Tech: Python, FastAPI, LangChain, React, TypeScript, PostgreSQL, Docker.

Lunar Crater Detection Pipeline | [GitHub](#)

Deep Learning

- End-to-end ML pipeline for automatic crater detection on LRO/Chandrayaan lunar imagery using CNN and FPN.

PUBLICATIONS, AWARDS & CERTIFICATIONS

Awards: Winner – NASA Space Apps Challenge 2025 (Global), Smart India Hackathon (SIH) 2025 Internal, PVG HackerHub 2025

Research Paper: Generative Architectural Design using Deep Learning (IJARSCT Journal, Dec 2024)

Certifications: IIRS-ISRO Certified: Geodata Processing using Python and Machine Learning (96% – A+)

Open Source: Published Python Packages "FaceParser" and "TrainIQ" on PyPI using BiSeNet architecture

GitHub: 89+ repositories | Achievements: Pair Extraordinaire, Pull Shark, YOLO

Community: Btechnotes – Academic notes platform with 100k+ B.Tech student users

LEADERSHIP & COMMUNITY

Founder & Developer – BTech Notes

2022 – Present

btechnotes.in – 100k+ B.Tech student users

- Founded free academic resource platform; crossed 1M+ views in first year using Webflow, Firebase, and Notion.

Student Council President

2023 – 2024

PVGCOE Nashik

- Organized technical events, workshops, hackathons, and tech fests; coordinated with faculty and industry speakers.